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Excel Formulas and Functions Exercise

Logical IF

Exercise

Write a formula using IF. Salary for an AVP will be $2,000 and for an analyst will be $1,000

Emp ID	Designation	Salary	Formula
E001	AVP	$2,000	=IF(B34="AVP", 2000,1000)
E002	Analyst	$1,000	=IF(B35="AVP", 2000,1000)
E003	Analyst	$1,000	=IF(B36="AVP", 2000,1000)
E004	Analyst	$1,000	=IF(B37="AVP", 2000,1000)

Logical AND

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales made by the employee should be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(AND(C36>50,D36="YES"),"YES","NO")
E002	Analyst	100	NO	NO	=IF(AND(C37>50,D37="YES"),"YES","NO")
E003	Analyst	49	YES	NO	=IF(AND(C38>50,D38="YES"),"YES","NO")
E004	Analyst	75	YES	YES	=IF(AND(C39>50,D39="YES"),"YES","NO")

Logical NESTED IF

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then convert the amount into dollars using Format

TABLE A	
Performance	Increment
A	$1,000
B	$500
C	$250

TABLE B				
Emp ID	Designation	Performance	Result	Formula
E001	AVP	A	$1000	=IF(F36="A","$1000",IF(F36="B","$500","$250"))
E002	Analyst	A	$1000	=IF(F37="A","$1000",IF(F37="B","$500","$250"))
E003	Analyst	B	$500	=IF(F38="A","$1000",IF(F38="B","$500","$250"))
E004	Analyst	C	$250	=IF(F39="A","$1000",IF(F39="B","$500","$250"))

LOGIC OR

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales made by the employee should be more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(OR(C38>50, D38="YES"), "YES","NO")
E002	Analyst	49	NO	NO	=IF(OR(C39>50, D39="YES"), "YES","NO")
E003	Analyst	49	YES	YES	=IF(OR(C40>50, D40="YES"), "YES","NO")
E004	Analyst	75	YES	YES	=IF(OR(C41>50, D41="YES"), "YES","NO")

LOGICAL IFERROR

Exercise

Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an error appears, then change it to ERROR IN SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary	FORMULA
E001	AVP	£80,000	£6,667	=IFERROR(C39/12, "ERROR IN SALARY")
E002	Analyst	£50,000	£4,167	=IFERROR(C40/12, "ERROR IN SALARY")
E003	Analyst	abcd	ERROR IN SALARY	=IFERROR(C41/12, "ERROR IN SALARY")
E004	Analyst	£40,000	£3,333	=IFERROR(C42/12, "ERROR IN SALARY")

Text - EXACT

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Result	Formula
TEST	Test	FALSE	=EXACT(A29,B29)
TEST	TEST	TRUE	=EXACT(A30,B30)
1234	1234	TRUE	=EXACT(A31,B31)
1.1	1.101	FALSE	=EXACT(A32,B32)

Text - TEXT

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results	Formula
21/05/2017	21/05/2017	21/05/2017	=TEXT(A34,"dd/mm/yyyy")
42876	21/05/2017	21/05/2017	=TEXT(A35,"dd/mm/yyyy")
12:00	12:00	12:00	=TEXT(A36,"hh:mm")
12345.78	£12,345.78	£12,345.78	=TEXT(A37,"£#,##0.00")
12345.78	£12,346	£12,346	=TEXT(A38,"£#,##0")

Text - CONCATENATE

Exercise

Join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned in the result column

String1	String2	String3	Result	USE	Formula
E001	AVP	800	E001-AVP-800	CONCATENATE	=CONCATENATE(A35,"-",B35,"-",C35)
E002	Analyst	500	E002Analyst500	CONCATENATE	=CONCATENATE(A36,B36,C36)
E003	Analyst	100	E003/Analyst-100	CONCATENATE	=CONCATENATE(A37,"/",B37,"-",C37)
E004	Analyst	400	E004:Analyst:400	&	=A38&":"&B38&":"&C38
E005	AVP	1000	E005 AVP 1000	&	=A39&" "&B39&" "&C39
E006	Analyst	250	E006 Analyst 250	&	=A40&" "&B40&" "&C40
E007	Analyst	500	E004;Analyst*500	&	=A38&";"&B38&"*"&C41

Text - Substitute

Exercise

Use SUBSTITUTE function to make changes to some of the strings below to get the results mentioned in the 'Result' column

String	Result	Formula
The dog is in the lawn	dog is in the lawn	=SUBSTITUTE(A39,"The","")
the dog is in the lawn	Dog is in lawn	=SUBSTITUTE(SUBSTITUTE(A40,"the"," "), "dog","Dog",1)
Dog	Fog	=SUBSTITUTE(A41,"D","F")
Curious Cat	Furious Cat	=SUBSTITUTE(A42,"Curious","Furious")

Text - VALUE

Exercise

Convert the values in Text column into numbers

Text	Result	Formula
$123.17	123	=VALUE(A22)
01/12/2017	43070	=VALUE(A23)
12/01/2017	42747	=VALUE(A24)

Text - UPPER

Exercise 1

Convert the values in Text column into UPPERCASE

Text	Formula	Formula
good	GOOD	=UPPER(A22)
how are you?	HOW ARE YOU?	=UPPER(A23)

Exercise 2

Use UPPER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result	Formula
match	MATCH	TRUE	=EXACT(UPPER(A28),UPPER(B28))
match	Match	TRUE	=EXACT(UPPER(A29),UPPER(B29))
correct	CORECT	FALSE	=EXACT(UPPER(A30),UPPER(B30))

Text - LOWER

Exercise 1

Convert the values in Text column into lowercase

Text	Formula	Formula
GOOD	good	=LOWER(A22)
HOW are you?	how are you?	=LOWER(A23)

Exercise 2

Use LOWER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result	Formula
MATCH	match	TRUE	=EXACT(LOWER(A28),LOWER(B28))
MATCH	mATCH	TRUE	=EXACT(LOWER(A29),LOWER(B29))
CORECT	correct	FALSE	=EXACT(LOWER(A30),LOWER(B30))

Text - LEN

Exercise 1

Find the length of the followig strings

Text	Result	Formula
Phone Number is +(0)1234	24	=LEN(A24)
This is used for data cleaning	30	=LEN(A25)